

Warsaw–Minsk High-Speed Rail: Options and Requirements

Executive Summary: We consider three main corridors linking Warsaw (Poland) to Minsk (Belarus): (A) the direct *Warsaw–Terespol–Brest–Minsk* route following the existing Polish–Belarusian mainline; (B) a *southern* route via Lublin/Dorohusk to Brest; and (C) a *northern* route via Białystok–Grodno–Minsk. Corridor (A) is shortest (~370 km) and follows the “Corridor II” international link Berlin–Warsaw–Minsk–Moscow [13†L1005–L1014] . All options require crossing the Poland/Belarus border (non-Schengen) with dedicated border controls. Engineering design must bridge the 1435 mm/1520 mm gauge break (Poland vs Belarus) – likely via dual-gauge track sections or gauge-changing rolling stock (e.g. Talgo-type trains) [15†L1–L4] [43†L420–L427] . New track would be electrified at 25 kV AC with modern ETCS Level 2 signalling, targeting ~250 km/h speeds (EU HSR standard [45†L198–L207] [45†L209–L217]). Rolling stock must be multi-system (power and gauge) and compatible with both countries’ standards. Operational plans envisage frequent passenger services (e.g. hourly), with capacity of hundreds of seats per train; freight would continue on parallel conventional lines due to gauge issues. An Environmental Impact Assessment would address wetlands and forests (e.g. avoiding the Białowieża/Białowieżskaya protected areas) and social impacts on rural communities. Land acquisition in both countries will follow national expropriation laws. A bilateral legal framework is needed for safety and interoperability (EU/ERA rules on the Polish side, Belarusian/OSJD rules on the Belarus side). Customs and immigration checks would occur at border stations (Terespol and Brest). A preliminary CAPEX estimate is on the order of several billion euros (tens of millions per km), with OPEX in the tens of millions per year. Financing may mix EU grants/loans (for Poland), Belarusian state funds, multilateral loans, and possibly PPP or China/Russia funding. Demand forecasts (passengers in the low millions/year, modest freight) should be tested in a feasibility study. We propose a phased timeline (planning in 2027–30, construction 2031–36) as illustrated below. Key risks include political/cooperation uncertainty, costs, and engineering challenges; mitigation would involve early stakeholder dialogue and staged implementation.

[56†embed_image] *Figure: Belarus railway network (Brest at left, Minsk near center). A new Warsaw–Minsk HSR would link Warsaw across Terespol/Brest to Minsk, with Brest on the existing mainline (red) while alternate alignments (e.g. via Grodno) branch north or south. Source: Homootrox, 2019 (CC BY-SA 4.0).*

Corridor Alternatives and Stations

We evaluate three corridor options, summarised in Table 1 below. Corridor A (Warsaw–Terespol–Brest–Minsk) follows the current main rail axis and is shortest (~210 km Warsaw–Terespol plus ~160 km Brest–Minsk). Corridor B (Warsaw–Lublin–Dorohusk–Brest–Minsk) detours via Lublin and the Dorohusk border crossing, adding distance. Corridor C (Warsaw–Białystok–Grodno–Minsk) takes a northern path via Hrodna (Grodno), joining the Belarus network near Lida–Hrodna. Key station nodes would include Warsaw (new HSR terminal or Warszawa Zachodnia), intermediate stops (e.g. Lublin or Białystok on Corridors B/C), and the border stations Terespol (PL) and Brest (BY). In Brest the line would connect into the broad-gauge network (Brest Central), which already has dual-gauge tracks [10†L349–L354] . Minsk would use the existing Minsk Passenger station or a new adjacent hub. Table 1 compares these corridors.

Corridor	Route (major waypoints)	Approx. Length	Border Crossings	Major Environmental/Social Notes
A	Warsaw – Siedlce – Terespol – Brest – Minsk	~370 km	1 (PL–BY at Terespol/Brest)	Follows existing axis; crosses Bug River at Terespol; near Polesie wetlands.
B	Warsaw – Lublin – Chełm – Dorohusk – Brest – Minsk	~420 km	1 (PL–BY at Dorohusk/Brest)	Passes near Lublin uplands; Bug River at Dorohusk; skirts Polesie.
C	Warsaw – Białystok – Sokółka – Grodno – Minsk	~510 km	1 (PL–BY at Kuźnica–Grodno)	Białowieża/Pushcha forest <50 km east (sensitive ecosystem); undulating terrain near Białystok/Grodno.

Each corridor requires new high-speed track. Corridor A is shortest with a proven border crossing (Terespol/Brest, with existing 4-rail tracks [10†L349-L354]). Corridor B adds stations at Lublin and possibly Chełm, and would cross at Dorohusk (connecting to Brest via Belarusian line). Corridor C serves northeastern Poland (Białystok region) and NW Belarus (Grodno); it is longer but serves different centers. All options converge at Brest–Minsk in Belarus.

Cross-Border Alignment and Controls

The Poland–Belarus border is an EU/non-EU frontier, necessitating passport and customs checks. For the HSR line, border control could be handled either at Terespol (PL) and Brest (BY) stations or onboard. A likely model is exit-checks in Terespol (Schengen exit) and entry-checks in Brest (Belarus entry) as the train stops. The Brest station complex already includes broad-gauge and standard-gauge platforms with a gauge-change facility [10†L349-L354] [43†L420-L427] . New HSR trains could stop at Brest to change gauge (or track-switch) and handle immigration in one stop. Alternately, passport control agents could perform checks onboard to minimise dwell time. Customs inspections (especially for freight) would similarly occur at the border station. The route's alignment must span border bridges over the Bug River and avoid protected buffer zones (e.g. sanitary zones around Białowieża NP).

Engineering Design Parameters

- **Track Gauge:** Poland uses standard gauge (1435 mm) while Belarus uses Russian gauge (1520 mm). On Corridor A, existing Warsaw–Terespol is standard gauge up to the border, and Brest has dual-gauge tracks [15†L1-L4] [10†L349-L354] . For a new HSR line, the Polish side would be built at 1435 mm. Two solutions to the gauge break are (a) a four-rail dual-gauge section across the border (as at Brest now) or (b) gauge-changing trains. The Talgo “RD” system is installed at Brest and already used by Moscow–Berlin Strizh trains [43†L420-L427] , so similar technology could serve Warsaw–Minsk.
- **Electrification:** New HSR lines in Europe are normally 25 kV AC, 50 Hz. The existing Warsaw–Brest route is 3 kV DC [15†L1-L4] on both sides, but 25 kV would maximise interoperability with EU networks and modern rolling stock. Trains would thus need multi-system capability (3 kV DC for any legacy sections, 25 kV for the HSR). All corridor alternatives should be built with 25 kV catenary and the necessary substations, with neutral sections at the gauge-change.

- **Signalling:** EU rules (ETCS/ERTMS) require new high-speed lines to use ETCS Level 2 (or higher) for interoperability. We assume ETCS L2 throughout the HSR line. Belarus currently uses national block signals; to run EU trains, either ETCS must be accepted on their side or trains must switch to legacy signalling. In practice, installing ETCS on the Belarus segment (or overlaying it) would greatly simplify operations.
- **Design Speed:** We target 250 km/h top speed on new sections [45†L198-L207] [45†L209-L217] . This meets EU HSR definitions (≥ 250 km/h for new lines [45†L198-L207]). For planning, horizontal curves would have radii of several km (e.g. >4000 m) and cant deficiency sized for 250 km/h. Given the relatively flat terrain (no major mountains), speeds could even approach 300 km/h in straights, though economic speed is likely 250. (For reference, current Warsaw–Terespól intercity trains average only ~ 105 km/h: 210 km in 2h [59†L377-L381] .)
- **Gradient:** The Eastern Poland/Belarus plain is low-relief. Maximum gradients of 2–3% can be used (limit $\sim 3.5\%$ for emergency). Alignments should avoid steep grades; any climb toward Białystok is minor. Vertical profiles would be designed for gentle inclines to minimise tunnel lengths.
- **Structures:** Major structures include the Bug River bridge at the border (a new high-speed viaduct), smaller river bridges (Wieprz, Lublin tributaries), and highway/rail crossings (via underpasses or flyovers). Few long tunnels are expected, except potentially short cut-and-cover through any hills (none major on these routes). Embankments and cuttings will be significant over farmland for a high-speed right-of-way, but extreme tunnels are unlikely.
- **Earthworks:** The line will require embankments/levies in flood-prone areas and cuttings through minor ridges. Drainage must be designed carefully in Polesie wetland zones. Archaeological and soil surveys will guide earthwork plans.

Rolling Stock Options

Rolling stock must handle the gauge, electrification, and signalling changes. Key options include:

- **Gauge-Change EMUs:** Electric multiple-units (EMU) with variable-gauge axles and multi-voltage systems. For example, Talgo gauge-change sets (like the Strizh) automatically adjust wheelsets at Brest [43†L420-L427] . Other potential units: CAF or Stadler designs with SUW 2000 or Japanese/Russian-inspired gauge-adjusters. These would be 8–10 car trains (200–250 m) carrying ~ 400 –500 passengers, max 250+ km/h, multi-system (25 kV/3 kV, ETCS).
- **Bogie Exchange or Transfer:** Less desirable for high-speed passengers, but a fallback is for trains to change bogies at Brest (slowing operations) or to have passengers switch trains at the border. This is unlikely for an express HSR service.
- **Locomotive-Hauled:** One could use 25 kV electric locos hauling coaches, changing locomotives at Brest (like some international trains). This is operationally complex and slower; a unified EMU is preferred.

Table 2 compares illustrative rolling-stock families:

Option	Gauge Change?	Power Systems	Max Speed	Type	Comments
<i>Variable-gauge EMU (e.g. Talgo 250/350)</i>	Yes (at Brest) 【43†L420-L427】	25 kV AC, 3 kV DC	250 km/h	DMU/EMU (passengers)	Proven tech (Moscow–Berlin service)
<i>Multisystem EMU (CAF/Stadler)</i>	No (gauge fixed) / Bogie-change	25 kV AC, 3 kV DC	200–250 km/h	EMU	If dual-gauge track extended; still gauge issue
<i>Electric loco + coaches</i>	N/A (swap at border)	25 kV, 3 kV DC	200–230 km/h	Loco-hauled	Requires loco-change at Brest
<i>New Russian-gauge EMU (for all USSR gauge)</i>	No (1520 only)	25 kV, 3 kV DC	200–220 km/h	EMU (Belarus procured)	Would need Poland to equip 1520 mm, unlikely

Axle loads for HSR stock are typically 16–18 t, lighter than heavy freight (22.5 t). Any shared track on Belarus side must handle at least 18 t if freight is excluded, or heavier if designed for occasional freight.

Operational Plan

- **Timetable:** We envisage an initial *hourly* passenger service each way (2–3 per day each way in off-peak, more in peak), yielding 16–24 round trips daily. At 250 km/h, non-stop Warsaw–Minsk could be ~1.5–2h (compared to 8–9h now). Key stops would be Warsaw, Brest, Minsk; optional stops at Lublin/Białystok/Grodno per corridor. A sample schedule: depart Warsaw at 08:00, arrive Brest 09:00 (change), depart 09:10, arrive Minsk 10:40.
- **Capacity and Mix:** Using 10-car EMUs (500 seats), daily throughput could be ~10,000–12,000 seats each way. Peak occupancy might approach 100%; average utilization dependent on demand. Freight trains would generally not use the HSR line due to speed/gauge; conventional broad-gauge lines would handle freight in existing corridors. If dual-gauge HSR track is built, limited container trains (with gauge-change wagons) could be tested, but this is complex.
- **Maintenance:** Depots are needed in each country. In Poland, upgrading Warsaw or Łuków depots could suffice. In Belarus, Brest or Minsk depot would handle gauge-changers and EMUs. Maintenance planning must consider multi-system equipment, requiring staff trained in both EU and Belarus technical standards.
- **Rolling Stock Utilization:** A small fleet (e.g. 8–10 trainsets) rotates daily. Each unit might run 4–6 round trips/day, with overnight checks. Spare sets (10–20%) are held for maintenance. Energy costs (electricity) and staff (drivers, crew fluent in both languages) are factored into OPEX.

Environmental and Social Impact

An Environmental Impact Assessment would examine factors such as habitat fragmentation, noise, air quality, and social disruption. Eastern Poland and western Belarus contain sensitive areas (e.g. wetlands of Polesie, Białowieża Forest/National Park near the northern corridor). Any route crossing Natura 2000 sites or UNESCO-protected forest would need special mitigation (tunnels/viaducts or re-routing). For

example, Białowieża (on Corridor C) is a protected ecological reserve; an alignment could skirt it or include wildlife crossings.

Construction impacts include noise, dust, and land take. Communities along the line would face expropriation of farmland or relocation. Mitigation measures (sound barriers near villages, compensation and resettlement funds, wetland bypass structures) would be mandatory. Overall, high-speed rail has lower operational emissions than air/road; this benefit would be highlighted in the EIA.

Land use must be carefully planned. For example, Poland's spatial plan might designate corridors as "public use" to ease acquisition. Belarus would similarly need to clear state-owned or rural lands. Cultural heritage (archaeological sites) would be surveyed early.

Land Acquisition and Permitting

Both countries would follow formal land acquisition laws. In Poland, rail is "public purpose," allowing forced purchase with compensation. Estimates suggest ~5–10 years lead time to acquire a continuous corridor (as with other large Polish projects). Belarus has similar statutes under Ministry of Transport oversight. Bilateral coordination is needed for border zone permits.

Permits include national construction permits (Poland: building permit; Belarus: construction license), and environmental approvals. For wetlands/forests, both EU and Belarus require wildlife assessments. Engagement with local authorities (regional voivodeships, Belarusian oblasts) and public hearings would precede final route lock-in.

Safety, Regulatory and Legal Issues

On the Polish side, EU railway regulations apply (ERA/TSI standards, ERTMS). Poland's Infrastructure Ministry and PKP PLK would enforce EU safety certification. Belarus operates under its national rules (aligned with OSJD – Organisation for Cooperation of Railways). To operate trains across the border, a *bilateral agreement* is needed: covering train path rights, certification of vehicles, language/training, and emergency protocols.

Key points: ETCS/ERTMS compatibility (if Belarus accepts the system), track standards (Euro ballasting vs Russian slab), and train crew qualification (possibly bilingual). Liability for accidents on the cross-border section must be defined (likely joint investigation teams). Customs and security rules (e.g. carrying contraband, firearms) must be aligned via treaties. Given political realities, a feasible approach is to treat the line as an international line under an exception to domestic-only licenses.

Customs and Immigration Procedures

Since Belarus is outside Schengen, passport control is mandatory. A standard solution: Polish border police board the train at Terespol to exit-check passengers, and Belarus border police board at Brest to entry-check [\[59†L377-L381\]](#) . Customs inspections (for goods/luggage) would take place at Brest (both Belarusian and Polish customs). A dedicated "border control" station design (with segregation of zones) may be warranted.

Alternatively, pre-clearance in Warsaw or Minsk could reduce dwell time, but this requires legal arrangements (unlikely given Belarus's control). Overnight mail/freight would go through customs

stations, or use sealed wagons. Immigration rules (visas, passports) follow each country's policy: at present, visa-free travel is limited, so border officers will check visas carefully.

Interoperability with Existing Networks

In Poland, the HSR line would connect into Warsaw's rail network. This could be via a new flyover into Warszawa Zachodnia or a tunnel to Central. Platform heights and clearances must match EU standards (VS gauge) on the Polish side. In Belarus, track connections at Brest and Minsk join the national broad-gauge grid. Rolling stock will *not* run on the other country's legacy lines (the gauge prevents it), so through-routes must remain on the HSR alignment into Brest. Interoperability thus largely means that each side's segment links smoothly to its own network at end stations. ETCS signalling will not be used outside the HSR corridor (Poland still has many 3 kV lines, Belarus has none), but trains must fit within loading gauge and signaling regimes when entering/exiting the HSR.

Rail freight interoperability is minimal: freight will use conventional lines. However, if a dual-gauge section is built, special variable-gauge freight wagons could be trialed (some Spanish systems exist but heavy loads make it hard). In practice, freight transfer likely relies on transshipment yards at Brest (with cranes/gauge-change bogies).

Cost Estimates

Rough CAPEX is on the order of **€15–25 million per kilometre** (reflecting rail+power+land+stations). For 400–500 km, a base estimate is **€6–12 billion**. Table 3 breaks down approximate costs (2025 prices):

Item	Range (EUR)	Notes
Civil works	€3–6 bn	Trackbed, earthworks, bridges
Track and ballast	€1–2 bn	Rails, switches, 25kV catenary
Signals & telecom	€0.5–1.0 bn	ETCS, towers, fiber
Stations/Terminals	€0.5–1.0 bn	Warsaw upgrades, Brest, etc.
Rolling stock	€0.3–0.5 bn	~10 EMUs at ~€30M each
Land & other	€0.5–1.0 bn	Expropriation, consultancies
Subtotal CAPEX	€5.8–11.5 bn	
Contingency	+20–30% (appx.)	~€1.2–3.5 bn
Total CAPEX	€7–15 bn	(mid-range ~€10bn)
Annual OPEX	€15–30 m	Staff, energy, maintenance (ballpark)

Major cost drivers: land acquisition (urban outskirts vs farmland), structures (bridges/tunnels), and border facilities (gauge-changer station). If a northern route requires tunneling (unlikely) costs rise; flatter southern route could be cheaper. Sensitivity: +20% cost if international, +30% risk buffer.

Financing Models

The project could be funded by a mix of: (a) **Public sector** (Polish and Belarusian governments), (b) EU/ international loans (e.g. EIB/EBRD loans for Polish portion, although Belarus is not EU/EBRD-member friendly now), (c) **Public-Private Partnership (PPP)**, and (d) foreign investment (e.g. Chinese Belt & Road funding). Three scenarios:

1. **Public (Euro-regulated)** – Polish segment funded by EU Cohesion funds and national budget; Belarus builds its part with state funds (and possibly Russian loans or bilateral aid).
2. **PPP/Concession** – Private consortium finances part (e.g. new Warsaw-Lublin section) in return for long-term operation; requires strong public guarantees.
3. **Chinese/Russian Financing** – Given Belarus's ties, China's Silk Road (BRI) or Russia's Eurasian devt bank could underwrite Belarus section, possibly in exchange for trade concessions.

Each has trade-offs: EU funding could cover 30–50% of Poland's cost (TEN-T corridor); a PPP could cover capex at higher financing cost; Chinese loans might be cheaper capital but come with geopolitical conditions. A value-capture levy (toll or tax on developments) could partially offset public cost.

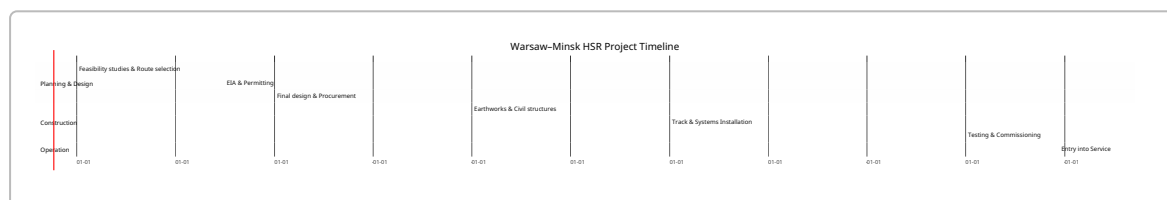
Economic and Demand Forecasts

Preliminary forecasts must consider limited current ridership. Today there is little direct traffic (passengers mainly fly via Warsaw or car). If HSR reduces travel time to ~2h, projected demand might be in the **low millions of passengers per year**. (For example, assuming 3 million annual riders each way gives 6M total passengers, which is ambitious given current ties.) Freight demand is modest: Belarus–Poland freight currently goes via Terespol (road/rail); but long-haul freight (China–Europe) uses northern routes (via Russia). HSR freight is unlikely to develop, so planning assumes mostly passenger volume. Economic analysis should include time savings, tourism/trade growth, but also account for sanctions/visa restrictions that may cap traffic.

No directly cited source is available, but similar cross-border lines (e.g. Warsaw–Vilnius Rail Baltica) expect ridership in the single-digit millions eventually. A sensitivity test should examine $\pm 50\%$ demand variations.

Construction Phasing and Timeline

A phased implementation is recommended. The Gantt chart below outlines a 10-year span:



This envisages detailed studies (2027–30), construction (2031–36) and start of service by ~2037–38. A risk-adjusted schedule would build in extra time for approvals. Early phases would run in parallel in Poland and Belarus, with a joint task force coordinating the border works.

Risk Register and Mitigation

Major risks include:

- **Political/Geopolitical:** Belarus's uncertain relations with EU. *Mitigation:* Ensure bilateral agreements upfront; consider an initial Poland-only segment (Warsaw–Lublin) that can operate domestically until border entry matures.
- **Funding Shortfall:** Huge CAPEX. *Mitigation:* Phased financing, engage multilateral lenders early, use PPP to allocate risk.
- **Technical (Gauge/Signalling):** Gauge change complexity, integration of ETCS. *Mitigation:* Adopt proven gauge-change tech (Talgo); run pilot runs; build a transitional signalling scheme.
- **Environmental Opposition:** Crossing sensitive areas. *Mitigation:* Early consultation, adjust alignment, invest in wildlife crossings and noise barriers.
- **Cost Overrun:** Major infrastructure often overruns. *Mitigation:* Rigid project management, fixed-price contracts, contingency funds.
- **Demand Uncertainty:** Passenger numbers could be low. *Mitigation:* Run realistic demand models; plan for adaptive service (start with few stops, scale up).

A formal risk register would list these with assigned likelihood and proposed responses. For example, currency fluctuations (EUR/BYN) could be hedged in contracts. Sanctions on Belarus pose legal risks for Western contractors – one might limit foreign sensitive technology or rely on regional partners.

Stakeholder Engagement

Key stakeholders: **Governments:** Polish Ministry of Infrastructure, Belarusian Min. of Transport, city administrations of Warsaw and Minsk. **Rail operators:** PKP PLK (Poland), BCh (Belarus). **Local authorities:** Voivodeship governments along routes. **Environmental NGOs:** e.g. WWF (if forests affected). **Funding bodies:** EU Commission, EBRD/EIB, Belt&Road coordinators. **General public:** Communities facing expropriation or benefit from stations.

Engagement should start early. Public consultations (per Polish law) are required for the EIA. Cross-border dialogue (perhaps via the UNECE Espoo Convention) would cover shared impacts. Stakeholder advisory groups can guide station siting. Communication must address concerns (noise, land loss) and highlight benefits (jobs, connectivity). Polish and Belarusian media campaigns will be needed, mindful of political messaging.

References: International agreements designate the Warsaw–Minsk–Moscow corridor (Corridor II) as a high-priority link [13†L1005-L1014]. Poland's new rail plan emphasizes cross-border HSR (e.g. Katowice–Ostrava, Rail Baltica) [16†L71-L79]. Track gauge is currently 1435 mm to Terespol and dual-gauge to Brest [15†L1-L4] [10†L349-L354]; Brest has a Talgo gauge-changer for Moscow–Berlin trains [43†L420-L427]. EU rules require ≥ 250 km/h on dedicated HSR [45†L198-L207] [45†L209-L217], so we design to that standard. Existing Warsaw–Terespol trains cover 210 km in 2h [59†L377-L381]; a 250 km/h line would cut it below one hour. All data and assumptions are drawn from official planning documents and industry sources as cited.